

1 Product Introduction

The device adopts an integrated handheld detection-and-countermeasure design with no external attachments. Its compact structure makes it easy for a single operator to carry. It supports 3 km omnidirectional detection and 2 km directional countermeasure against DIY FPV drones and conventional UAVs in the surrounding airspace. Equipped with SDR jamming technology and an auxiliary aiming system, it can perform long-duration 360° direction finding, parsing, and positioning of UAVs, and achieve comprehensive suppression to force UAVs to crash, hover, land, or return.



Figure 1 Portable Integrated Detection-and-Countermeasure Counter-UAS Device

2 Key Features

- Detection and Countermeasure Integration

Integrates UAV detection and jamming technologies, enabling 3 km omnidirectional detection and 2 km directional countermeasure.

- Adaptive Jamming

Uses dynamically adaptive dedicated jamming codes to jam DIY FPV drones and conventional UAVs, efficiently blocking UAV flight-control, data/video transmission, navigation, and other signals.

- Full-Band Jamming

Covers common UAV frequency bands used domestically and overseas, and supports arbitrary frequency-band configuration to meet various defense requirements.

- Custom Jamming

Equipped with wideband spectrum detection and supports manual frequency tuning for jamming to counter non-standard UAVs.

- Auxiliary Positioning

Equipped with an auxiliary aiming system that can display target information in real time, aim at target positions, and synchronize pilot location information, etc.

Basic Specifications	
Detection Frequency Band	100 MHz-6000 MHz
Detectable Models	DIY FPV drones (ELRS/TBS Crossfire and other protocols), conventional UAVs (DJI/Autel/Parrot, etc.), and Wi-Fi UAVs
Detection Range	≥3 km (open environment), 0.5-1 km (urban environment)
Direction-Finding Accuracy	10° (open environment)
Jamming Frequency Band	400 MHz / 800-900 MHz / 1.2 GHz / 1.4 GHz / 1.5 GHz / 2.4 GHz / 5.2 GHz / 5.8 GHz / 700 MHz-3 GHz software-customized jamming channel
Jamming Angle	Directional jamming; horizontal 45°, elevation 30°

Jamming Range	≥ 2 km (J/S ratio 2:1); DJI Mini 4 Pro/Mavic 3 ≥ 1 km (J/S ratio 1:1); DIY FPV (ELRS/TBS Crossfire protocol)
Jamming Response Time	<5 s
Screen Display	5-inch touch screen displays self-test status, UAV model, jamming frequency band, date/time, battery level, jamming duration, operation logs, etc.
Battery Type	Replaceable lithium battery
Single-Battery Endurance	≥ 15 min (full-band transmission enabled)
Overall Dimensions	653 × 407 × 187 mm ±5 mm
Total Weight	<7 kg
Protection Rating	IP54
Operating Temperature	-30°C to +55°C